MMCOE

Unit Test

Subject: Data Structures and Algorithms | Date: 13 October 2026

Duration: 1 hour | Total marks: 20

Instructions

Attempt ONE alternative from each unit. Each alternative carries the full unit marks.

Answer Q1 or Q2, and Q3 or Q4. Draw neat labelled diagrams where required.

Q1. Unit 1 — 10 marks

(a) Explain binary search trees and illustrate the insertion of keys 40, 20, 60, 10 and 30. [5]

(b) Analyze the time complexity of searching a balanced binary search tree. [5]

OR

Q2. Unit 1 — 10 marks

(a) Compare arrays and linked lists with respect to insertion, deletion and memory use. [5]

(b) Write an algorithm to reverse a singly linked list and state its complexity. [5]

Q3. Unit 2 — 10 marks

(a) Explain the data flow in the given client-server architecture. [5]

(b) Design a queue using two stacks and justify the amortized time complexity. [5]

OR

Q4. Unit 2 — 10 marks

(a) Evaluate the recurrence T(n) = 2T(n/2) + n and state its asymptotic bound. [5]

(b) Explain Dijkstra’s algorithm and identify the condition required for edge weights. [5]